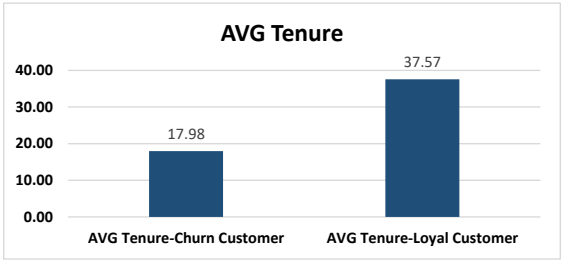
EDA-Telco-Customer-Analysis

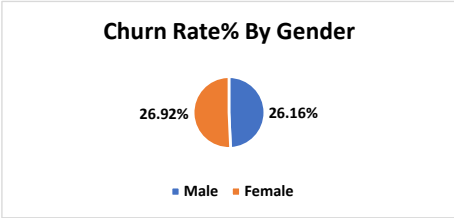
Key Statistics

Metric	Count/Value
Total Cutomers	7043
Churn Customers	1869
Churn Rate%	26.54%
AVG Tenure-Churn Customer	17.98
AVG Tenure-Loyal Customer	37.57



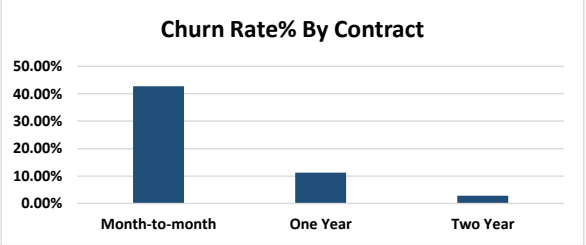
Churn By Gender

Gender	Total Customers	Churn Customer	Loyal Customer	Churn Rate%
Male	3555	930	2625	26.16%
Female	3488	939	2549	26.92%
Gerand Total	**7043**	**1869**	**5174**	**26.54%**



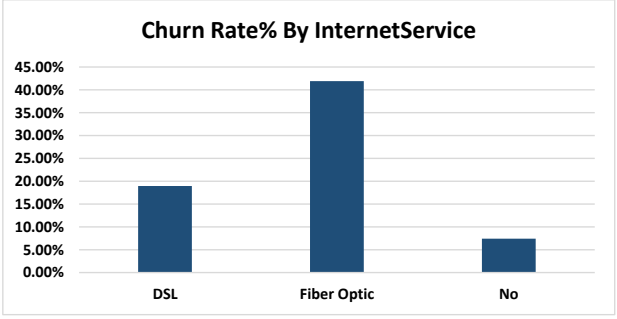
Churn BY Contract

Contract	Total Customers	Churn Customer	Churn Rate%
Month-to-month	3875	1655	42.71%
One Year	1473	166	11.27%
Two Year	1695	48	2.83%



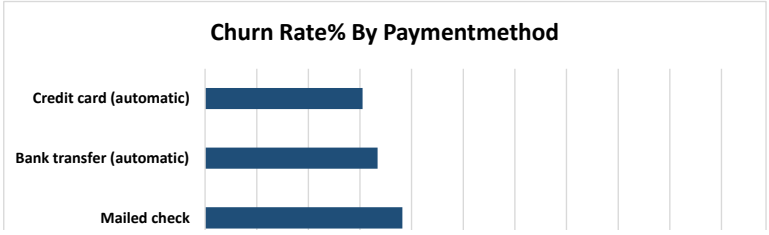
Churn By InternetService

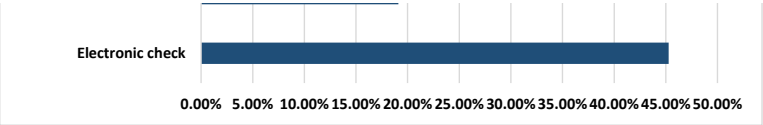
InternetService	Total Customers	Churn Customer	Churn Rate%
DSL	2421	459	18.96%
Fiber Optic	3096	1297	41.89%
No	1526	113	7.40%



Churn By Paymentmethod

Paymentmethod	Total Customers	Churn Customer	Churn Rate%
Electronic check	2365	1071	45.29%
Mailed check	1612	308	19.11%
Bank transfer (automatic)	1544	258	16.71%
Credit card (automatic)	1522	232	15.24%





Key Insights & Actionable Recomendation

1) Overall Churn & Critical Tenure Gap

Out of **7,043** total customers, **1,869** customers churned, resulting in a **26.54%** churn rate. Churned customers have a significantly lower average tenure **(17.98 months)** compared to loyal customers **(37.57 months).**
Recommendation: Focus retention efforts on the first **18 months** of the customer journey through proactive onboarding and early-stage loyalty programs.

2) Contract Type Risk
Month-to-month contracts drive the highest churn at **42.71% (1,655 out of 3,875** customers), whereas **2-year** contracts drop churn to just **2.83% (48 out of 1,695** customers).
Recommendation: Incentivize month-to-month customers to switch to long-term plans by offering initial contract discounts or exclusive perks.

3) Fiber Optic Service Dissatisfaction
Fiber Optic internet users experience a high churn rate of **41.89% (1,297 out of 3,096** customers), compared to DSL users at **18.96% (459 out of 2,421** customers).
Recommendation: Review Fiber Optic service quality, network reliability, and pricing to resolve underlying satisfaction issues.

4) Payment Method Influence
Electronic Check users show the highest churn rate at **45.29% (1,071 out of 2,365** customers), while Automatic Payment methods (Bank/Credit Card) keep churn low at **15%–17% (490** total churned customers).
Recommendation: Encourage customers to enroll in Auto-Pay by offering small monthly bill credits or fee waivers.

5) Gender Neutrality in Customer Churn
Gender has almost no impact on churn, with female churn at **26.92%** (**939 out of 3,488** customers) and male churn at **26.16% (930 out of 3,555** customers).
Recommendation: Direct marketing and retention budgets toward contract length and payment method optimizations rather than gender-based targeting.